

Automated Driving Car using RPI Board and CNN Algorithm

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Abstract

Self-driving vehicles will provide more than a luxury in this era. In this study, we are implementing an upcoming technology in Machine learning and neural networks called Autonomous cars or self-driven cars. It provides relief to humans from travelling stress and helps reduce traffic congestion and accidents. This study produces a successful model that navigates through the traffic and tackles all the obstacles placed in its path. This self drive car is implemented using Machine Learning algorithms and Convolutional neural networks in a way that provides complete automation to the driver. This car performs the following functions: self-parks, avoids traffic congestion, follows lane markings, avoids obstacles and detects vehicles and moves accordingly based on its distance from other vehicles. Previously, designs were made in such a way that all these functions were implemented in separate vehicles. But this car design integrates all these functions in a single model which increases its effectiveness, efficiency and reduces the cost of building it separately.

Keywords: Autonomous car, CNN, image processing, ultrasonic sensor

INTRODUCTION

Driving has become an integral part of people's life, since they rush around trying to get their errands done. The accidents and errors encountered while driving due to various factors like use of cell phones, traffic problems and carelessness are few of the major issues to be tackled. Since the number of accidents is increasing rapidly over time, necessary measures have to be taken in order to overcome these human errors and help mankind. To overcome this problem, self-driving cars or autonomous car was introduced which drives to the destination given without any human help. Autonomous cars or self driven cars provide humans with a luxury that lessen the commuting burden. It will eliminate accidents caused due to negligence of drivers, intoxication and various other factors. It provides comfort to those who have to commute long distances without facing the hurdles of traffic.

The main self-governing vehicle "Linrrican Wonder" was presented in 1925, which was a radio-controlled driverless vehicle. It went on New York City lanes, from Broadway down to Fifth Avenue. In 1926, the Linrrican Wonder sent Radio driving forces through radio receiving wires which got the radio motivations and sent the sign to circuit-breakers which worked little electric engines that direct every development of the vehicle. Yet, first real independent and genuinely self-sufficient autos were created during the 1980s with the assistance of Carnegie Mellon University and ALV extends in 1984 and a short time later in 1987 by Budeswer University Munich's EUREKA Prometheus Project and Mercedes-Benz.

Autonomous car aims at travelling from one point to the destination without any human interfering and to mainly avoid the